December 11, 2023

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:
Bionic energy management - 2

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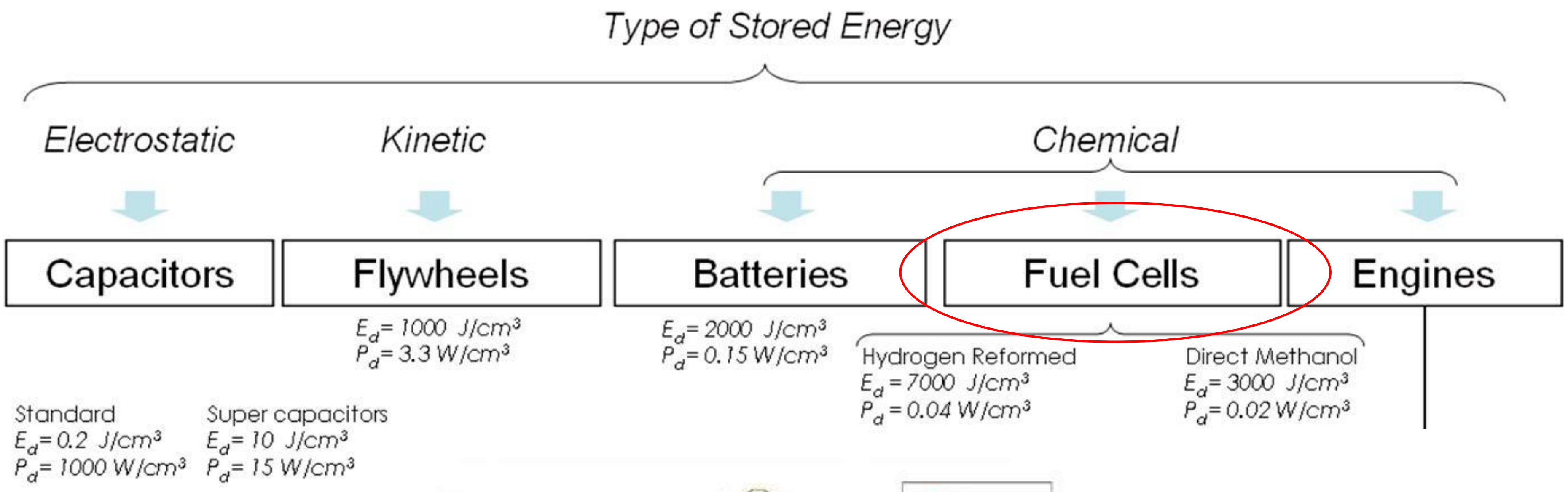
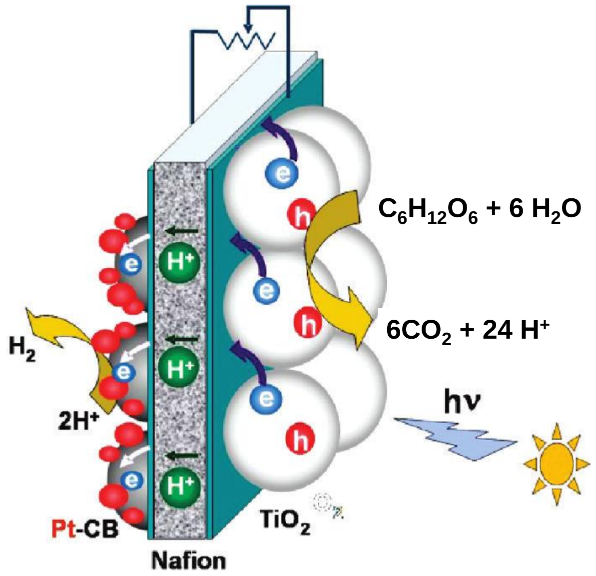
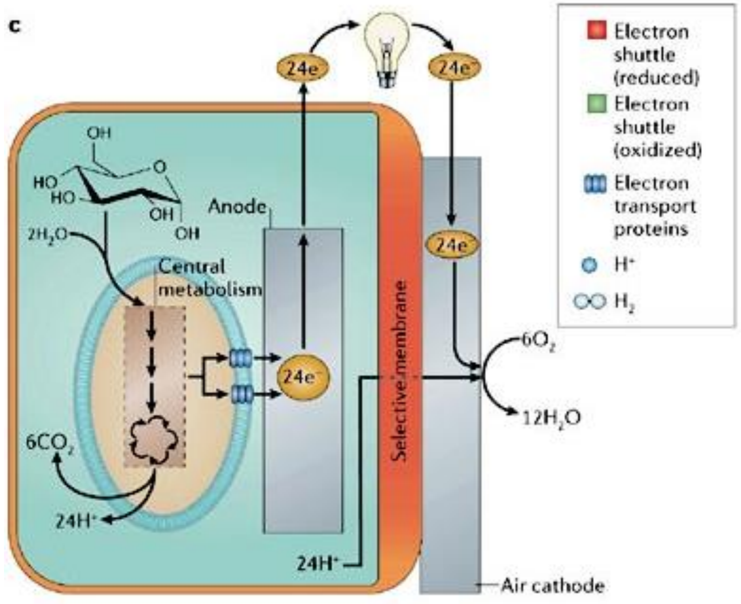
THE BIOROBOTICS INSTITUTE

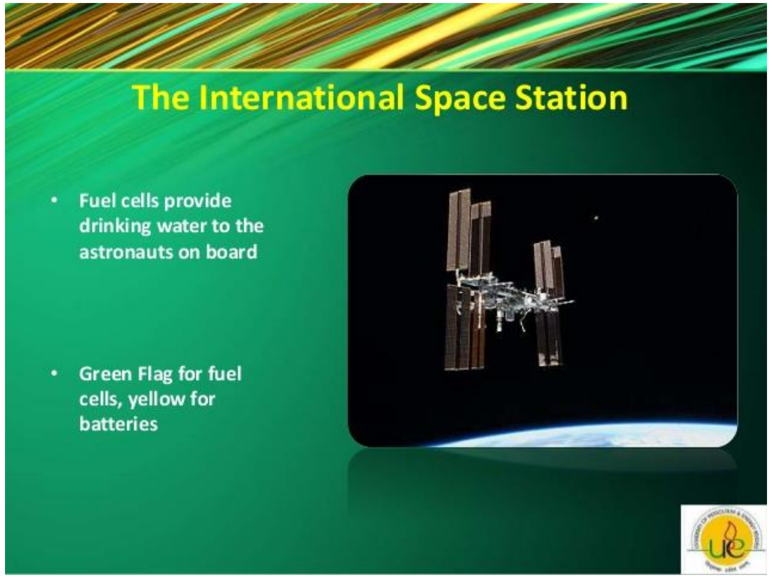
Scuola Superiore Sant'Anna

High capacity and efficient in on-board storage of energy



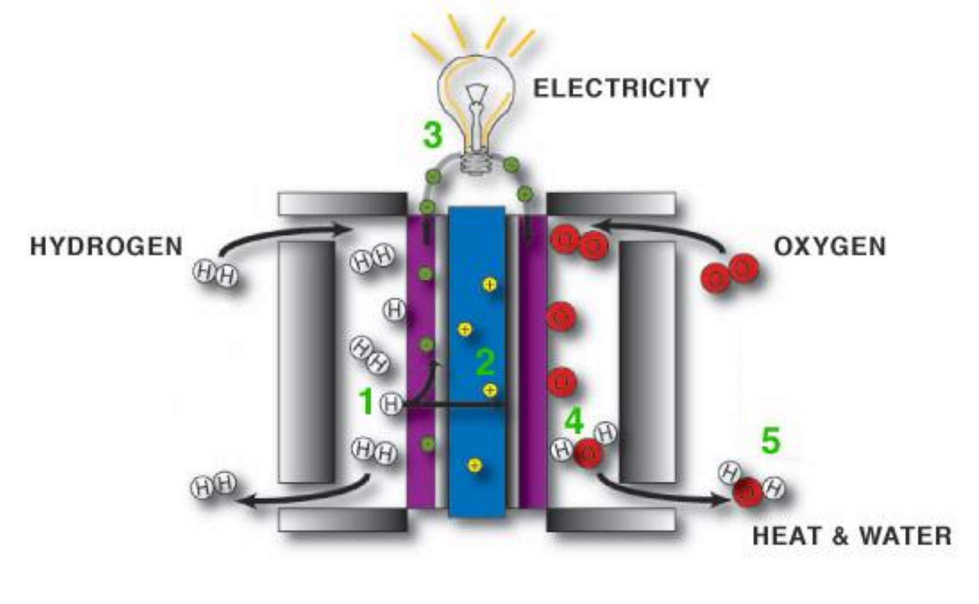






Fuel Cells

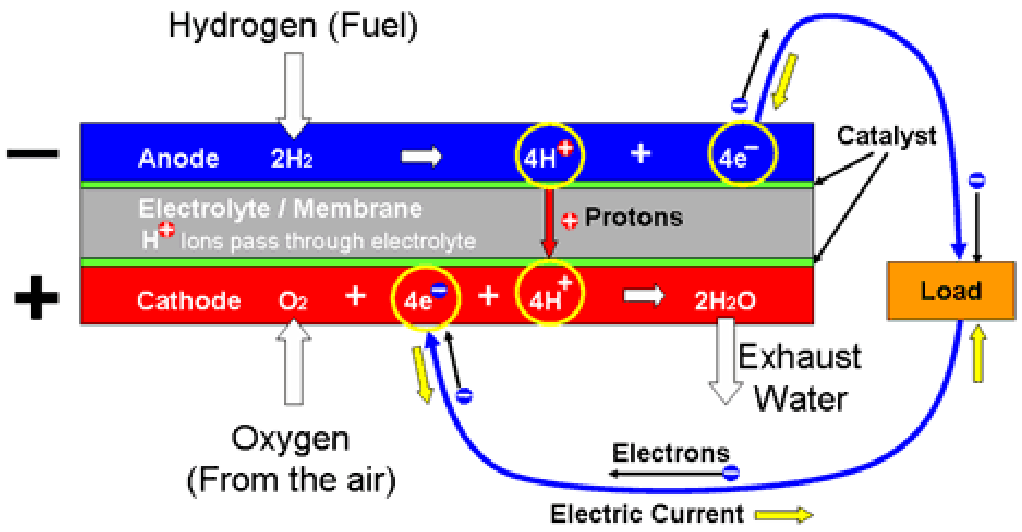
Electrochemical cells that convert chemical energy into electricity through an electrochemical reaction of hydrogen fuel with oxygen or another oxidizing agent.



Fuel cells don't store energy like batteries. They only provide electrical energy while the active chemicals are supplied to the electrodes. They can produce electricity continuously for as long as fuel and oxygen are supplied.

Fuel Cells

Proton Exchange Membrane (PEM) Fuel Cell



Fuel Cell Electrochemistry

the electrons travel round the external circuit to the cathode

Electrodes: porous carbon which allows the active gases to pass through and the electrode surfaces support platinum catalysts.

Electrolyte: thin, sheet of acidic, solid organic polymer impermeable to electrons.

Fuel Cells

Direct Methanol Fuel Cells (DMFC)

Methanol is directly used instead of hydrogen into the fuel cell by eliminating the reforming process

Anode Reaction:

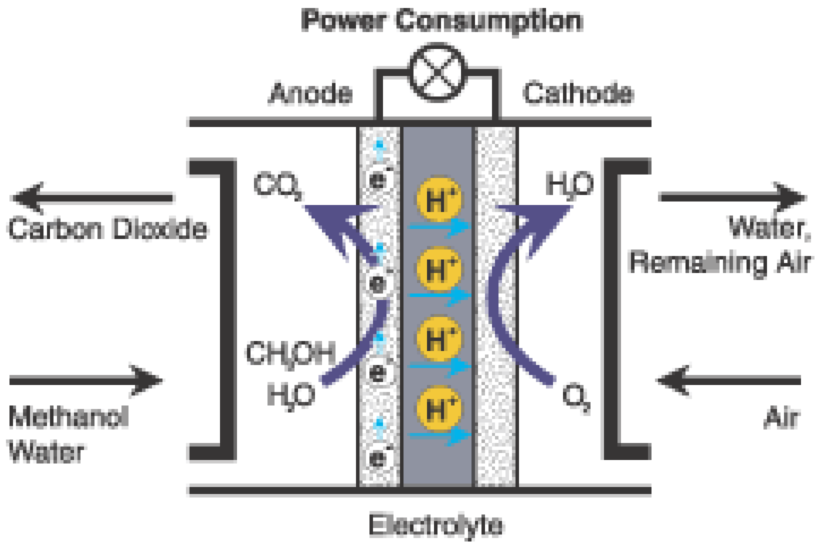
$CH_3OH + H_2O \Rightarrow CO_2 + 6H^+ + 6e^{--}$

Cathode Reaction:

$3/2\ O_2 + 6H^+ + 6e^{--} \Rightarrow 3H_2O$

Overall Cell Reaction:

$2CH_3OH + 3O_2 \Rightarrow CO_2 + 4H_2O$

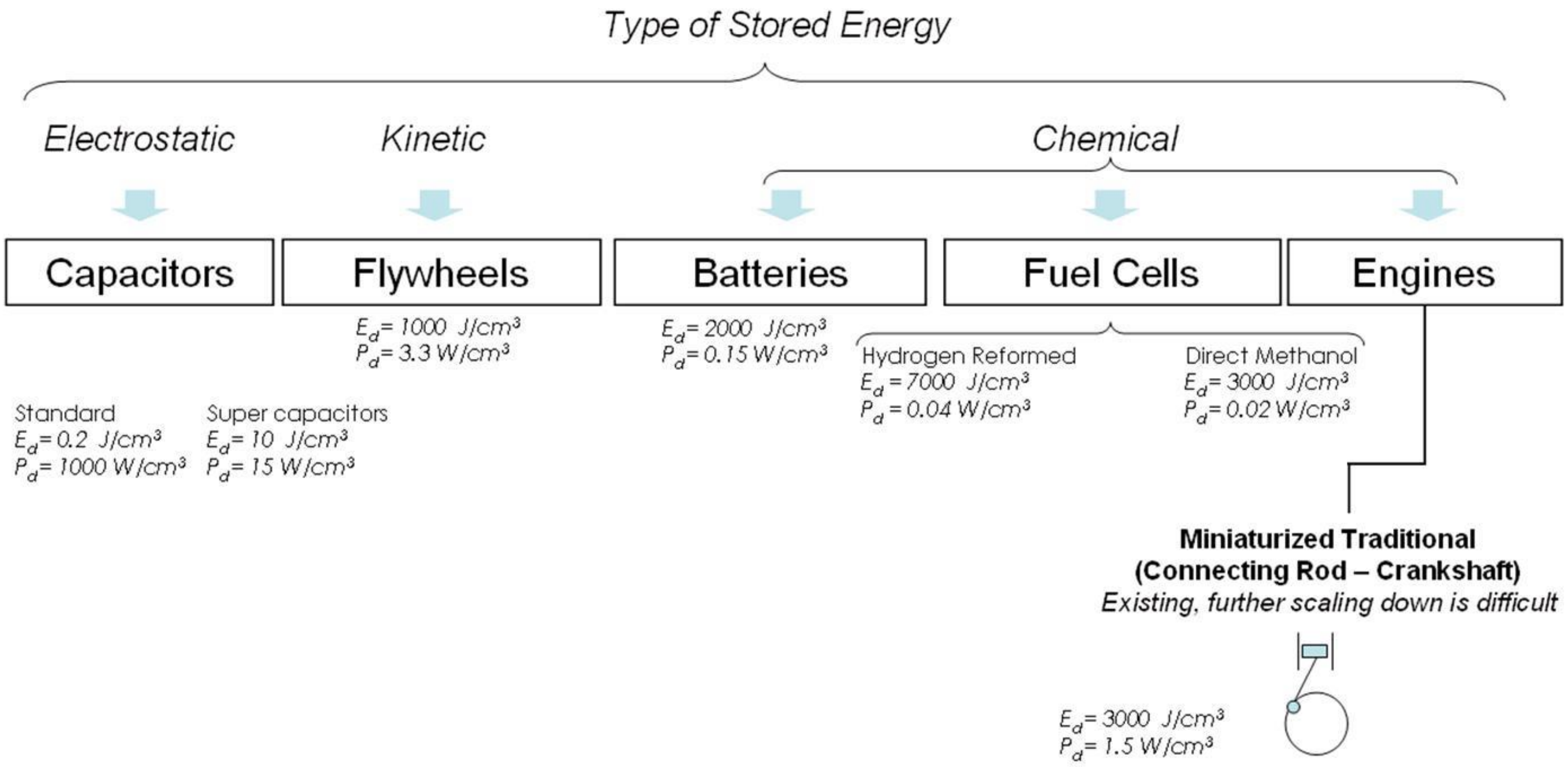


Specific content of chemical energy: 6 kWh/kg

Operating temperatures: 50°C to 120°C

Power outputs: between 25 W and 5 kW

High capacity in on-board storage of energy



Alternative energy sources

Bionspired robots should be **energetically autonomous** and therefore will be equipped with an on-board power generating system able to:

- **Extract** energy from the environment
- **Store** it in a suitable form, and
- **Use** it efficiently so that mission goals can be achieved in a timely manner

Ideally, bioinspired robots should be able to "survive" on naturally occurring "food" by **mimicking metabolic systems** of natural organisms so as to metabolize organic or inorganic substrates into electricity and/or storable fuels.

Examples of nature-inspired options that are currently under investigation in this direction include:

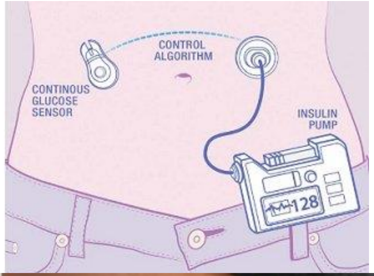
- **Photosynthetic** systems based on photo(electro)catalytic processes
- **Photo-Fuel Cells**
- **Microbial** and **enzymatic** fuel cells
- **Direct glucose PEM or alkaline** fuel cells

Energy issues in wearable and implanted artificial organs

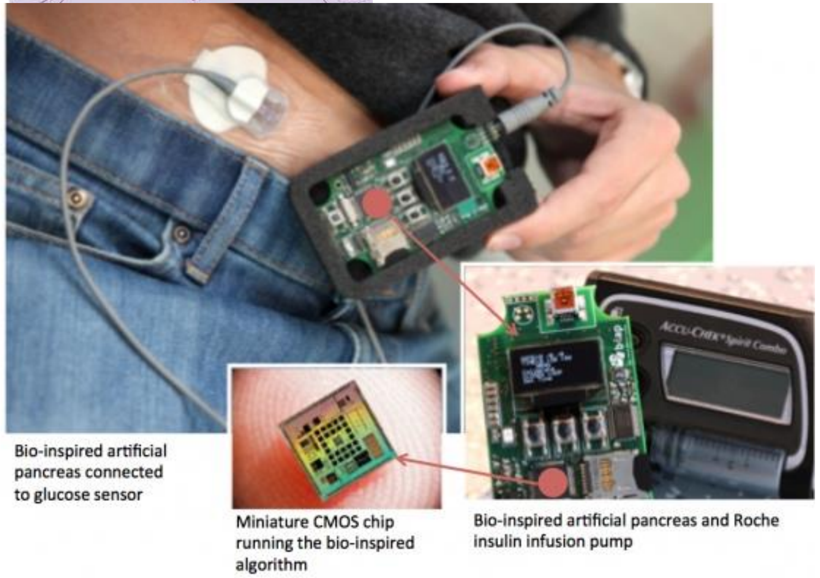
WEARABLE ARTIFICIAL ORGANS:

The need of wearing bulky energy sources limits the patients' quality of life

Artificial Heart
(Carmat, 2013 - powered by an external lithium-ion battery pack*)*



Artificial Pancreas
(Dexcom-Roche, 2013 – externally powered glucose sensor + insulin pump*)*

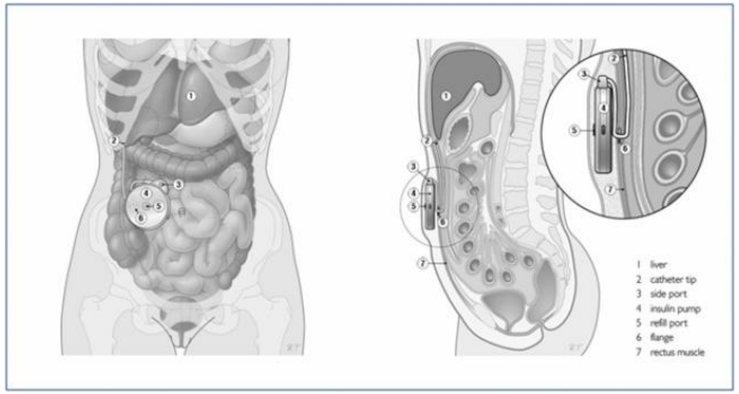


Energy issues in wearable and implanted artificial organs

IMPLANTED ARTIFICIAL ORGANS:

The limited battery lifetime implies the need of periodical surgical interventions to replace batteries

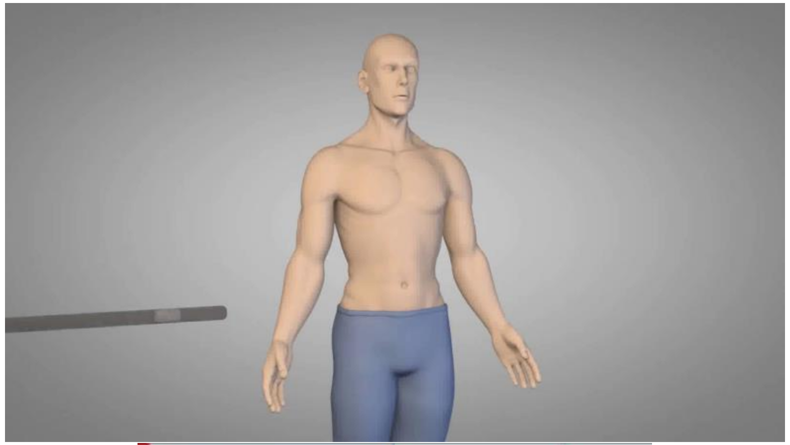
No fully autonomous implanted artificial hearts are available at present
(energy demand is too high to allow a fully implantable system)

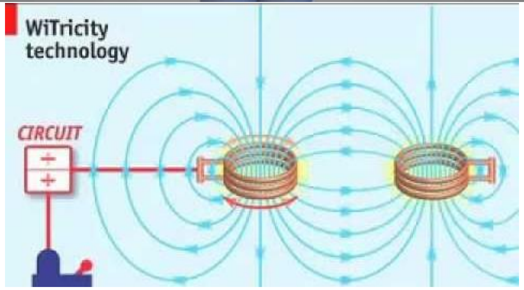


Fully implanted Artificial Pancreas
*Minimed 2007®, by Medtronic – **retired from the market** in 2011, due to poor acceptability: frequent surgical interventions needed, for insulin refilling (~ every 40 days) and battery replacing (~ every 1 year)*

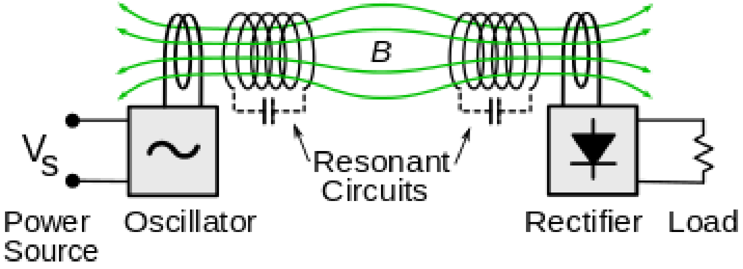
Energy issues in wearable and implanted artificial organs

Wireless energy transfer is emerging as a powerful tool to recharge implanted batteries in a non-invasive way





- **Transcutaneous Energy Transfer (TET)**
- **Free-range Resonant Electrical Energy Delivery (FREE-D)**



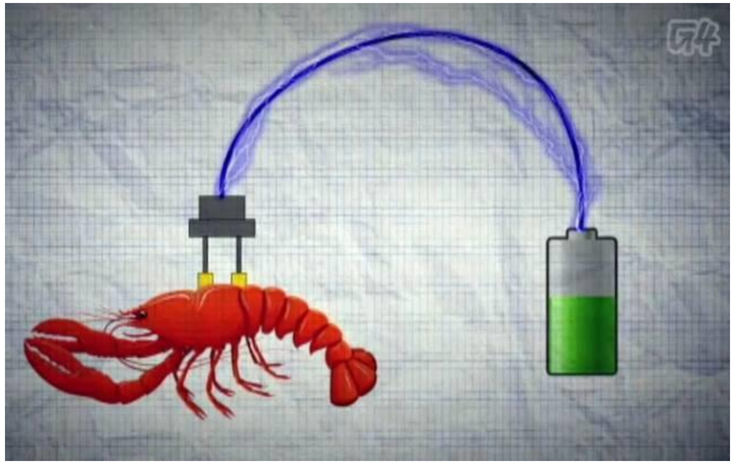
Strategies based on electromagnetic coupling of internal and external coils

Implantable biofuel cells harvesting energy from the internal physiological resources

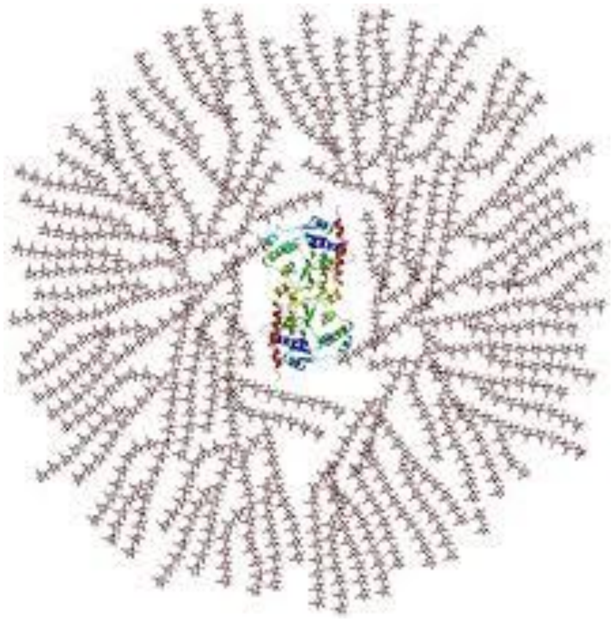
Biofuel cells converting chemical energy into electricity upon biocatalyzed chemical reactions have recently emerged as promising alternative sources of sustainable electrical energy. Implantable biofuel cells suggested as micro-power sources operating in living organisms are still overlooked and very challenging to design within bioelectronic systems. Enzyme-modified electrodes implanted in living organisms resulted in biocatalytic oxidation of **glucose** and reduction of **oxygen** in the biofluid inside a body thus producing electrical power from the biological source. Integration of the “electrified” organism with the biofuel cells allows for the activation of an electronic device (e. g. digital watch) as a model electronic device.

Implantable biofuel cells harvesting energy from the internal physiological resources

Future implantable medical devices such as cardiac defibrillators/pacemakers, deep brain neurostimulators, spinal cord stimulators, gastric stimulators, foot drop implants, cochlear implants, insulin pumps, *etc.* powered by implanted biofuel cells extracting electrical energy directly from a human body are possible, resulting in bionic human hybrids.



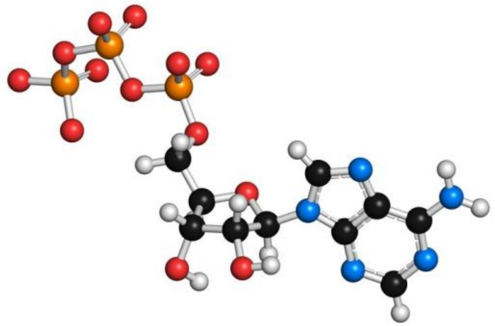
Energy storage in living organisms

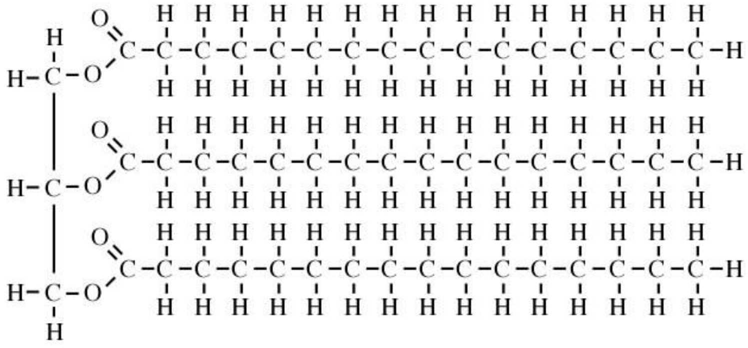


Living organisms have two major types of energy storage:

- stored energy in the form of covalent chemical bonds

- stored energy in the form of gradients of charged ions across cell membranes





Energy storage in living organisms

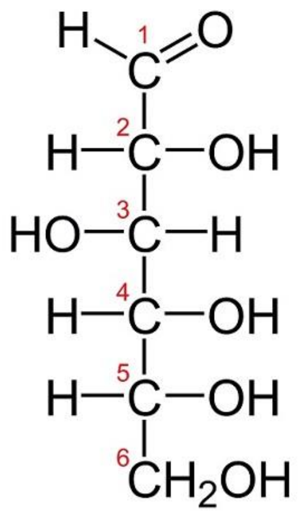
Carbohydrates:

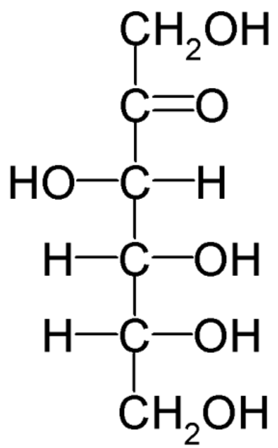
-Organic chemical compounds consisting of **carbon**, **hydrogen** and **oxygen** atoms.

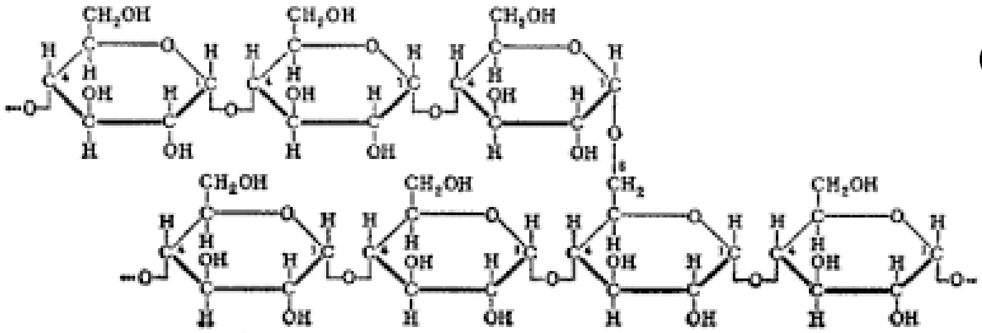
- Numerous function in living organisms including **energy storage**

- Monosaccharides are **aldehydes** or **ketones** with many **hydroxyl** groups

-Monosaccharides can be linked together forming **polysaccharides**







Energy storage in living organisms

Functional groups

Oxidrylic

Aldehyde

Carbonyl

Carboxy

Amine

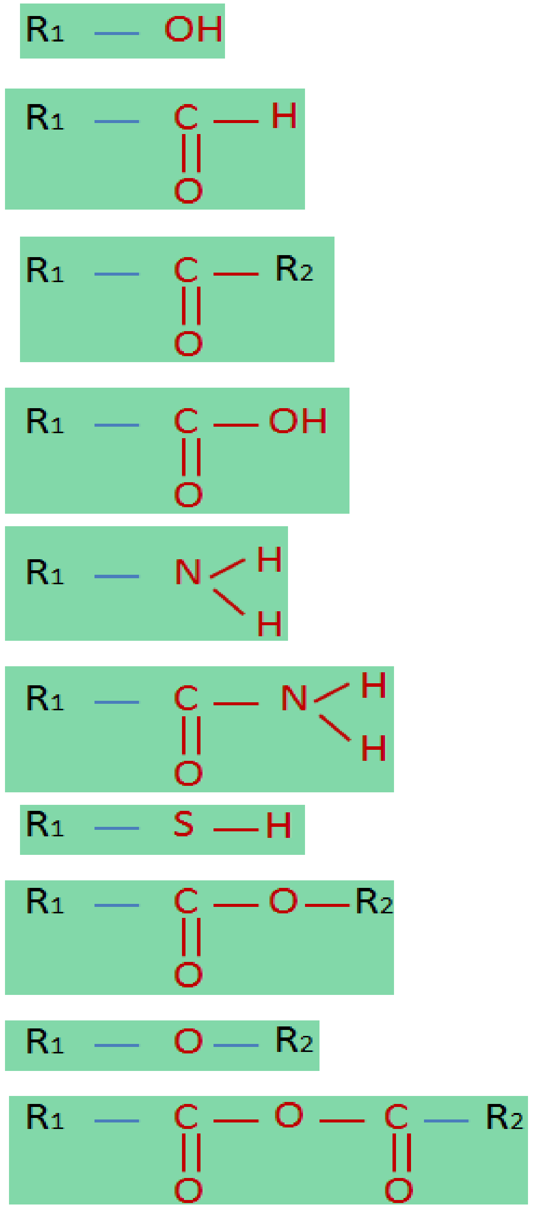
Amide

Thiol

Ester

Ether

Anhydride



Energy storage in living organisms

Monosaccharides:

- The major source of fuel for metabolisms

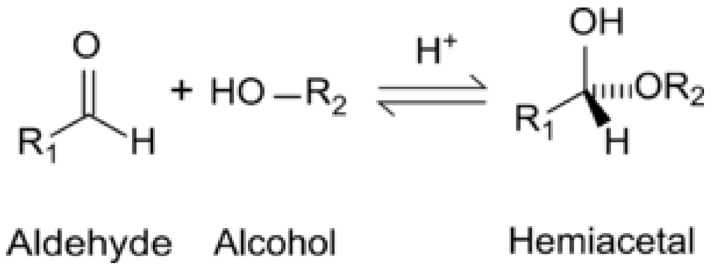
-Monosaccharides are used both as an energy source (glucose being the most important in nature) and in anabolism

-When monosaccharides are not immediately needed by many cells they are often converted to more space-efficient forms, often polysaccharides

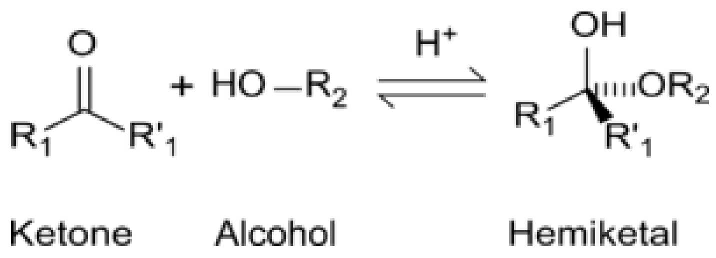
Energy storage in living organisms

Monosaccharides can exist in both a straight-chain and ring form:

The aldehyde/ketone carbon (C=O) reacts with **oxidrylic group** (–OH) forming a **hemiacetal/hemiketal** with a new C–O–C bridge



Formation of hemiacetals



Formation of hemiketals

Energy storage in living organisms

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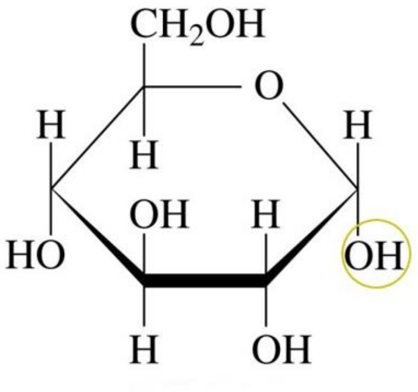
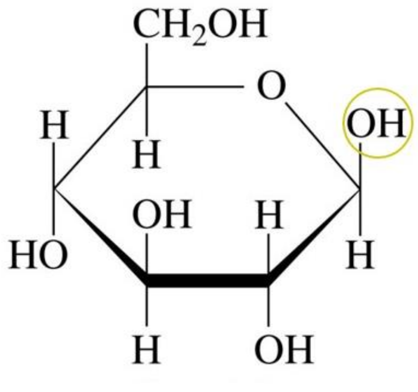
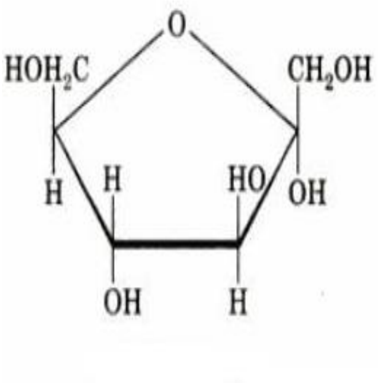


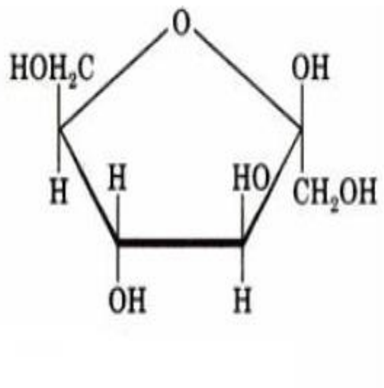
α-Glucose



ß-Glucose



α-Fructose



ß-Fructose

Glucose anomers

Fructose anomers

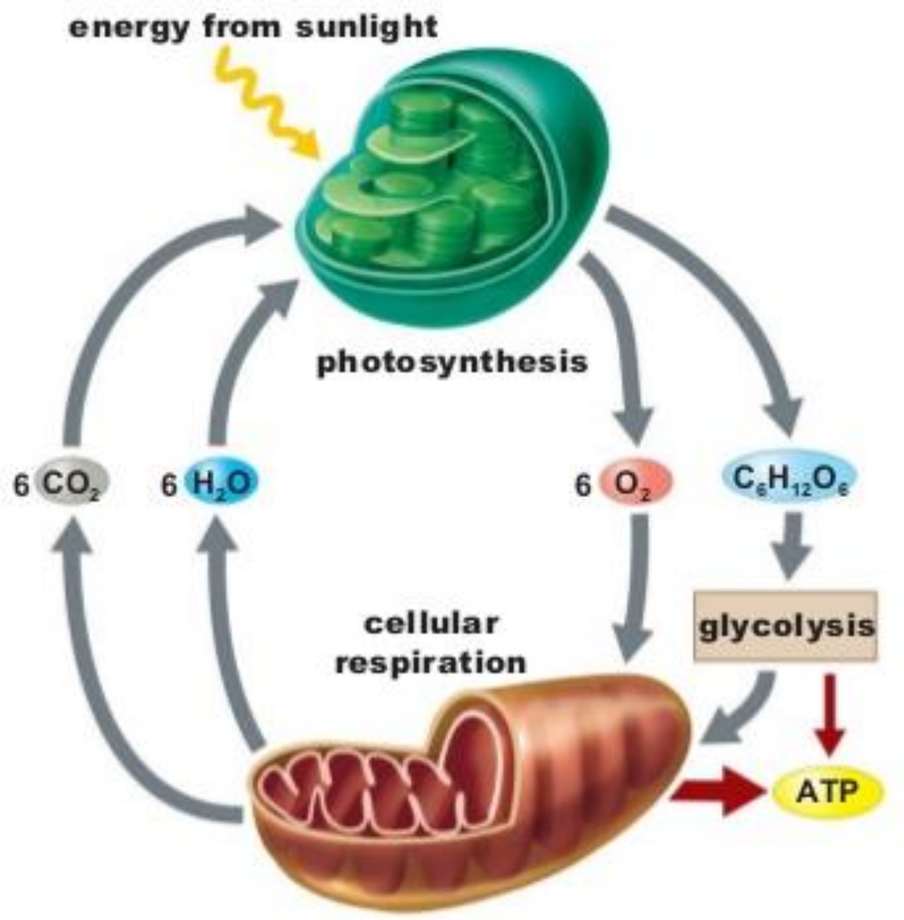
Energy storage in living organisms

Glucose is a key energy storaging molecule

Glucose is involved in two complementary processes :

- energy + water + carbon dioxide = glucose + oxygen

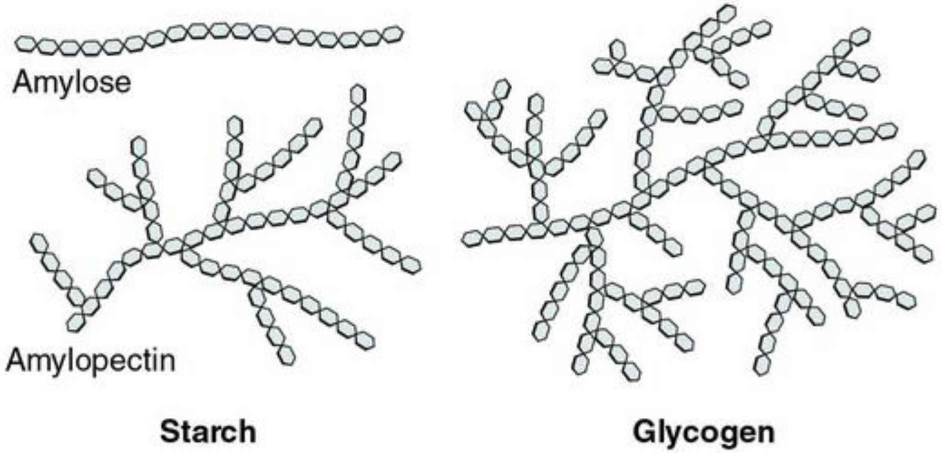
- glucose + oxygen = energy + water + carbon dioxide



Energy storage in living organisms

Glucose is stored as glycogen predominantly in liver and muscle cells in animals.

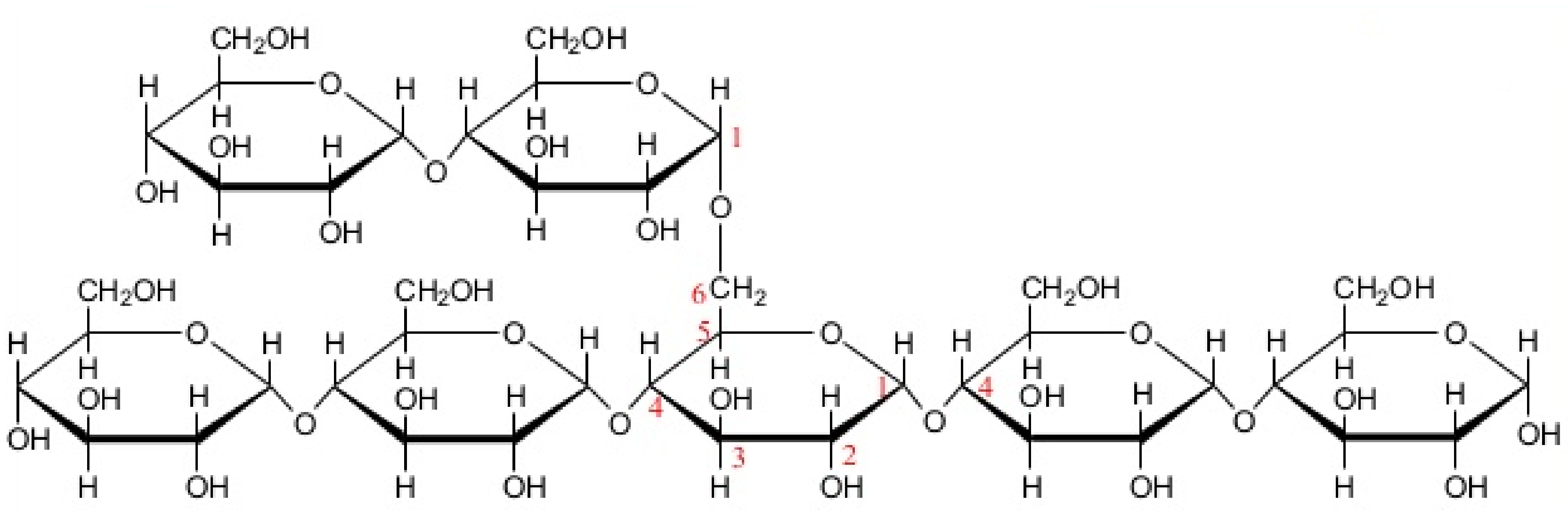
Glycogen is a highly-branced polymer of glucose units.



Its basic structure is similar to that of amylopectin, but with only about 8 to 12 glucose units between branch points (n = 4 to 6).

It can be broken down quickly to glucose.

Energy storage in living organisms



Glycogen is a polymer of glucose residues linked by

-α(1-4) glycosidic bonds, mainly

-α(1-6) glycosidic bonds, at branch points

Energy storage in living organisms

The strong affinity of carbohydrates for water makes their storage in large amounts inefficient (large molecular weight of the solvated water-carbohydrate complex).

Excess carbohydrates are catabolised to form Acetyl-CoA and then they enter the **fatty acid synthesis pathway**.

Lipids are commonly used for **long-term energy storage**.

Energy storage in living organisms

Lipids

a group of hydrophobic or amphiphilic substances including different categories of molecules that comprise important energy storage molecules (e. g. acid fats, triglycerides).

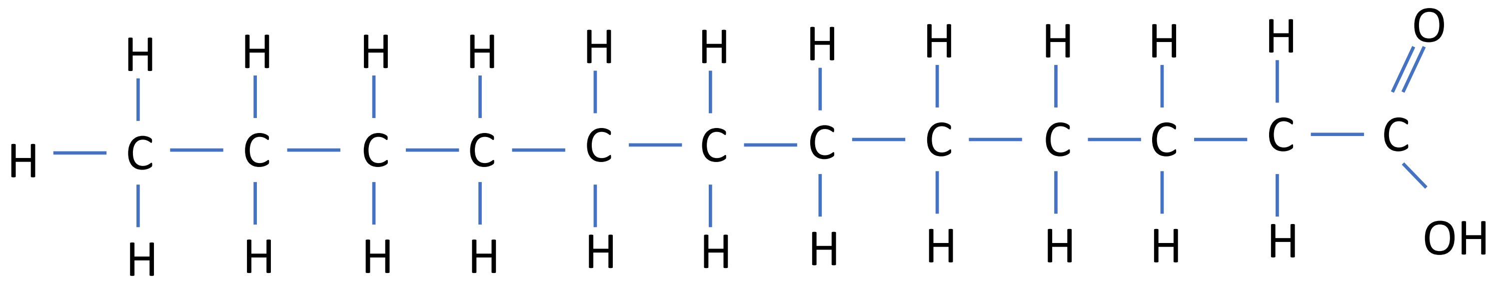
complete oxidation of fatty acids → 9 kcal/g

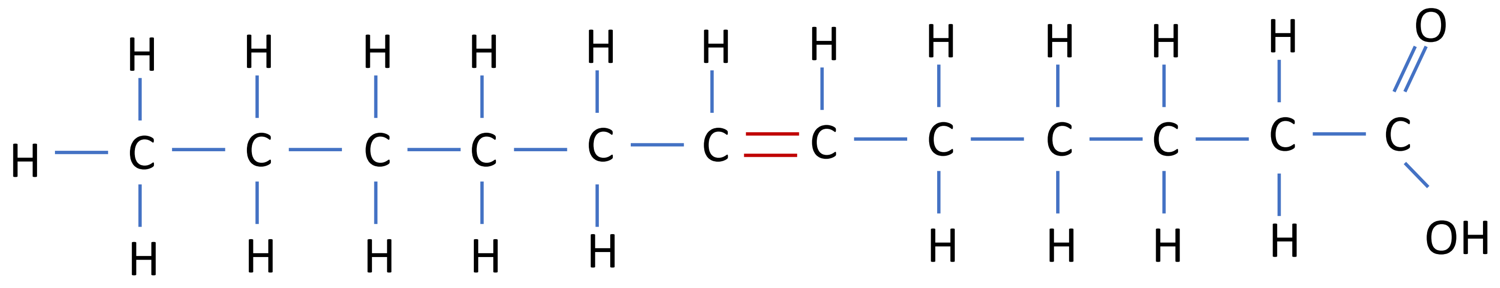
complete oxidation of carbohydrates → 4 kcal/g

Energy storage in living organisms

Fatty acids

Carboxilic acids with a long aliphatic chain (satured or unsatured)





Energy storage in living organisms

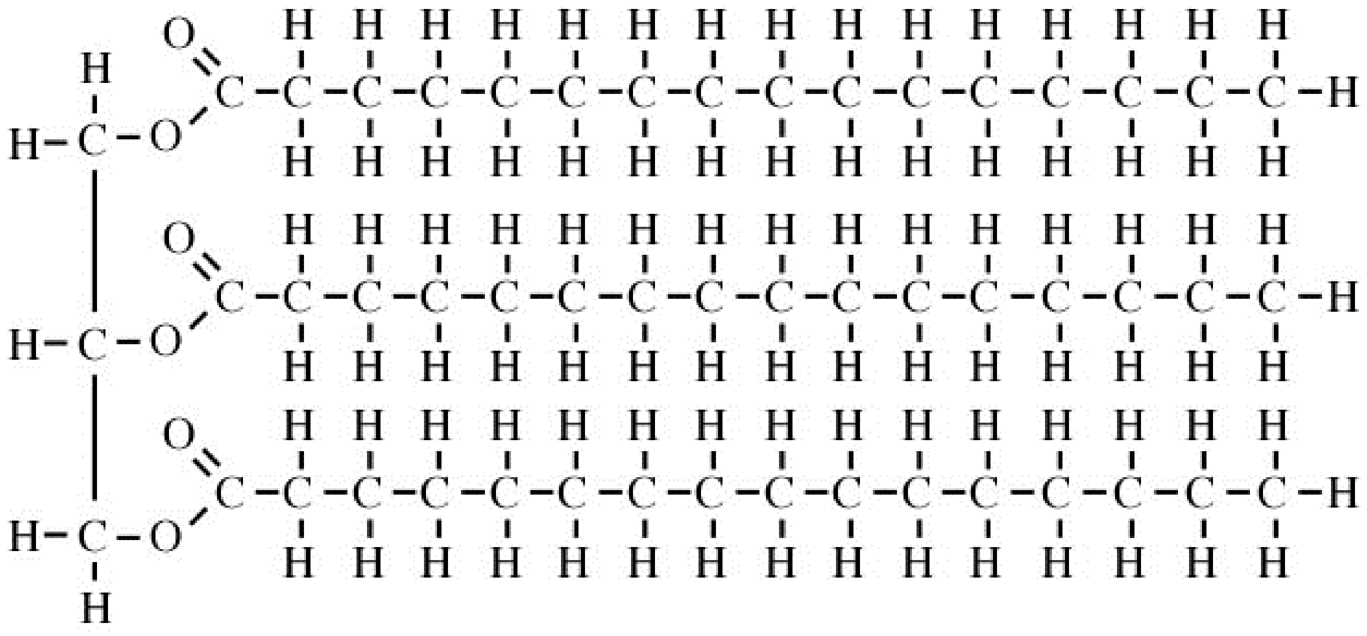
Fatty acids

Chemical Names and Descriptions of some Common Fatty Acids				
Common Name	Carbon Atoms	Double Bonds	Scientific Name	Sources
Butyric acid	4	0	butanoic acid	butterfat
Caproic Acid	6	0	hexanoic acid	butterfat
Caprylic Acid	8	0	octanoic acid	coconut oil
Capric Acid	10	0	decanoic acid	coconut oil
Lauric Acid	12	0	dodecanoic acid	coconut oil
Myristic Acid	14	0	tetradecanoic acid	palm kernel oil
Palmitic Acid	16	0	hexadecanoic acid	palm oil
Palmitoleic Acid	16	1	9-hexadecenoic acid	animal fats
Stearic Acid	18	0	octadecanoic acid	animal fats
Oleic Acid	18	1	9-octadecenoic acid	olive oil
Ricinoleic acid	18	1	12-hydroxy-9-octadecenoic acid	castor oil
Vaccenic Acid	18	1	11-octadecenoic acid	butterfat
Linoleic Acid	18	2	9,12-octadecadienoic acid	grape seed oil
Alpha-Linolenic Acid (ALA)	18	3	9,12,15-octadecatrienoic acid	flaxseed (linseed) oil
Gamma-Linolenic Acid (GLA)	18	3	6,9,12-octadecatrienoic acid	borage oil

Energy storage in living organisms

Triglyceride

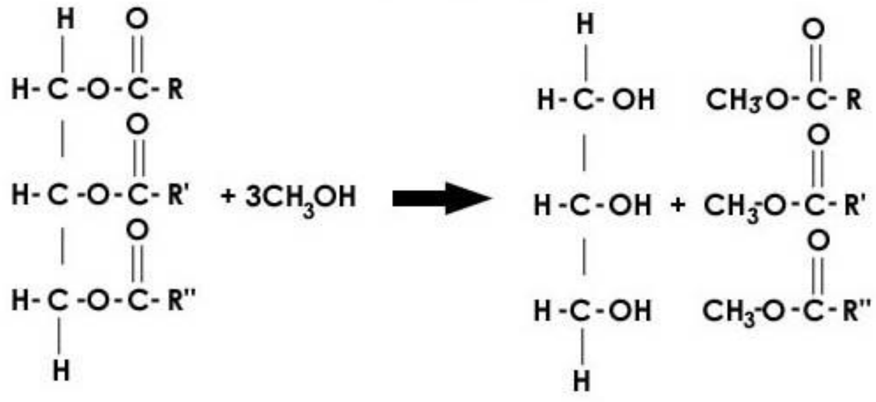
Glycerol and three fatty acids forming an ester.
They are the main constituents of body fat in animals and vegetable fat.



Energy storage in living organisms

Triglyceride

Triglycerides are also split into their components via transesterification and the resulting fatty acid esters can be used as biodiesel.

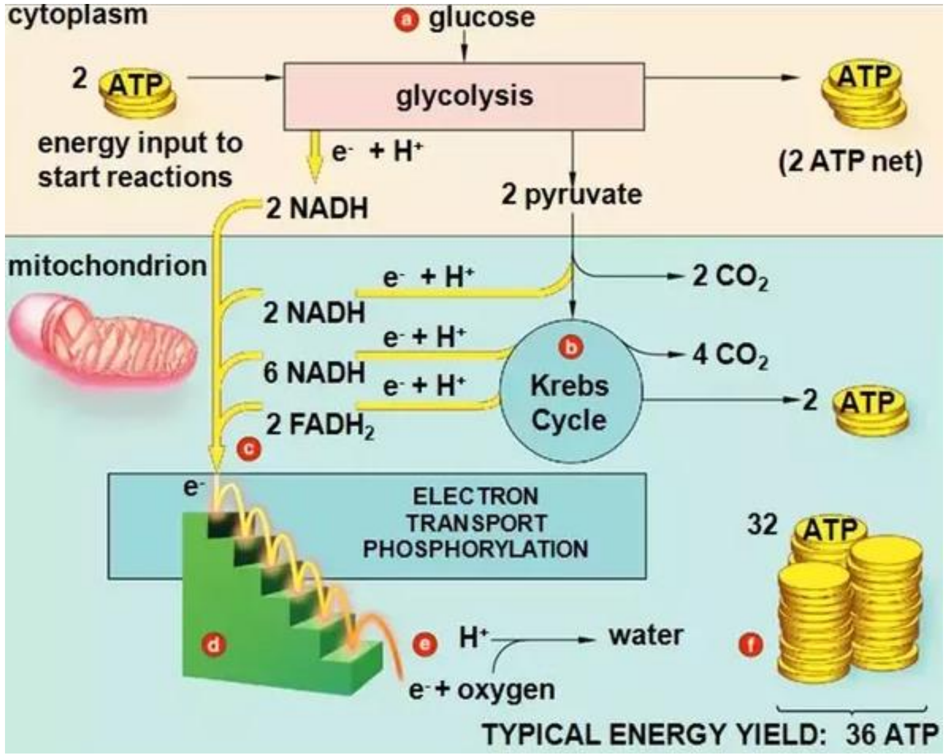


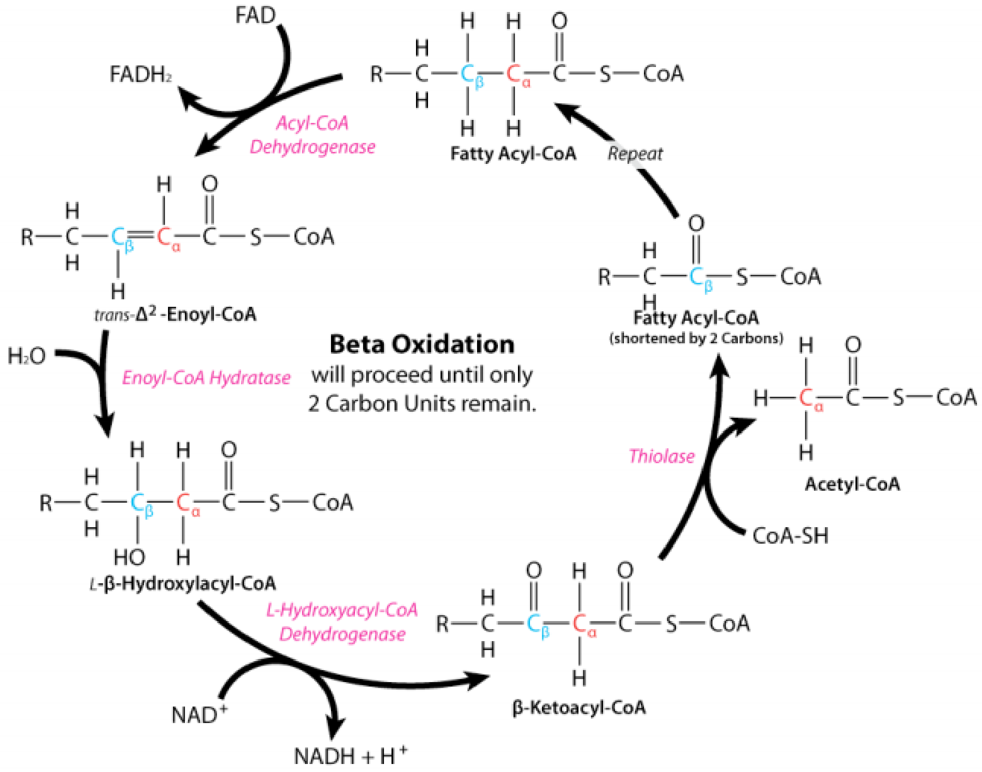
Triglyceride Methyl alcohol Glycerol Methyl esters

Energy storage in living organisms

Complete oxidation of Glucose = 36/38 ATP

Complete oxidation of fat acids (e. g. C_{20}) = 165 ATP



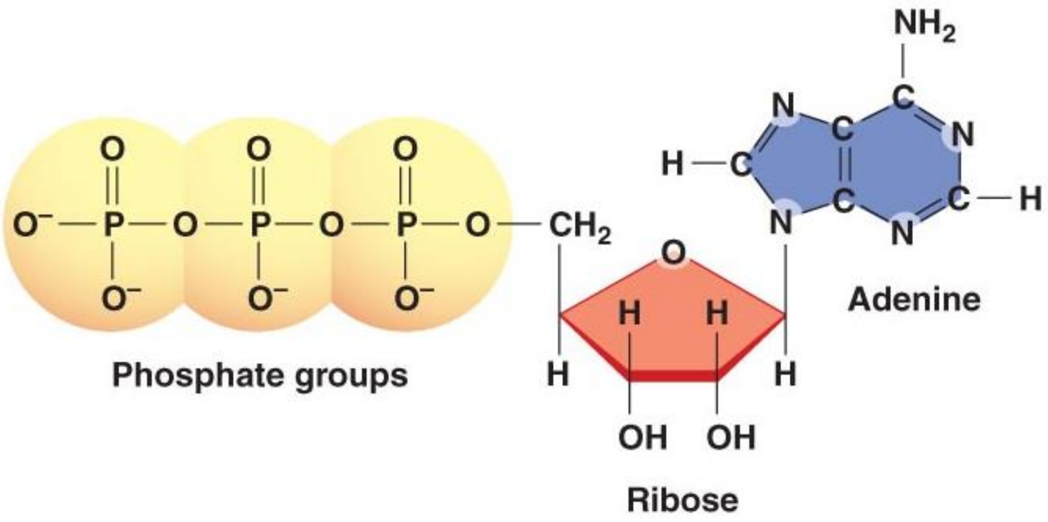


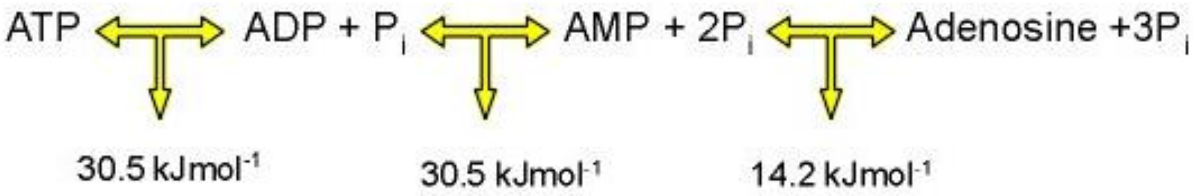
Energy storage in living organisms

Adenosine triphosphate (ATP)

energy storage for short periods

Need to store energy in more complex molecules

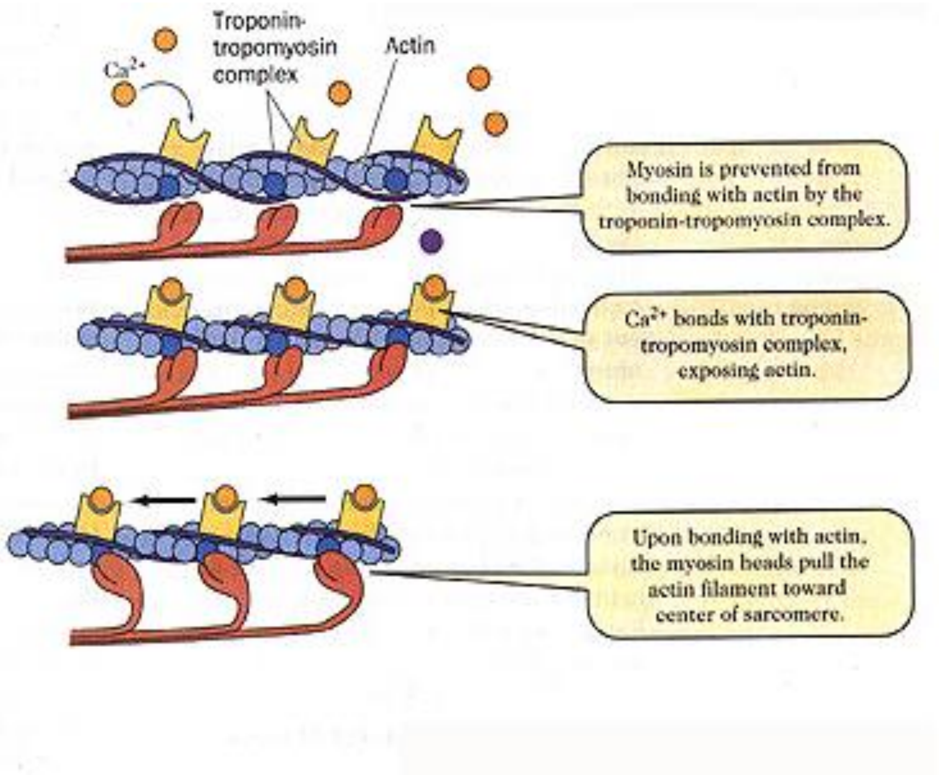


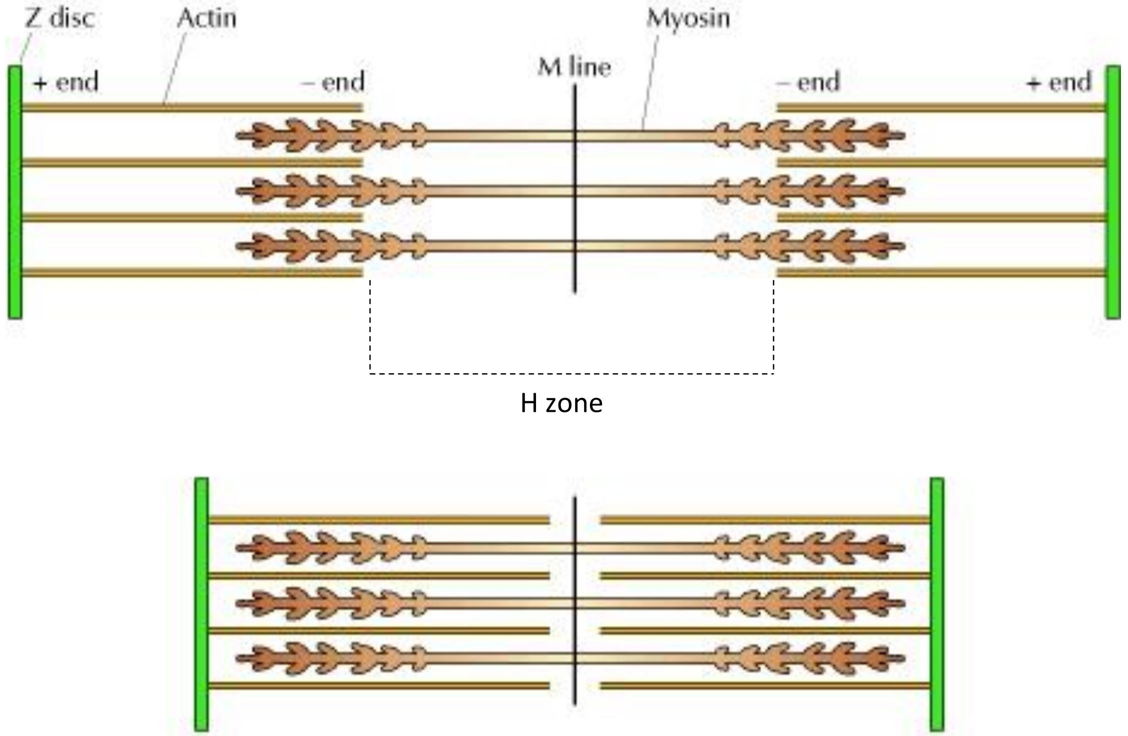


Converting chemical energy into mechanical energy in animals

Muscles

the oxidation of carbohydrates and fats (but also by anaerobic chemical reactions) produces ATP to power the movement of the myosin heads.





Converting chemical energy into mechanical energy in animals

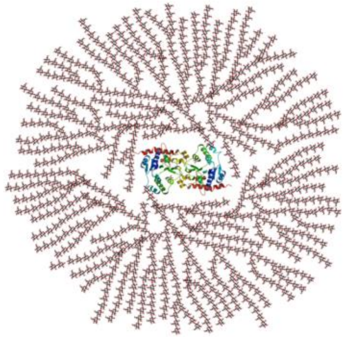
Muscles

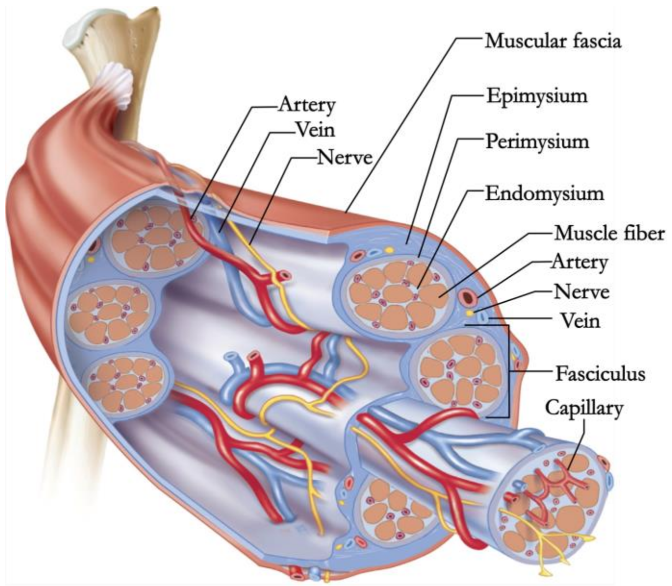
Creatine phosphate: short-term store of energy which is generated from ATP and can regenerate ATP with creatine kinase.

ATP ⇄ ADP + Pi

creatine ⇄ creatine phosphate

Glycogen: rapidly converted to glucose when energy is required for sustained, powerful contractions.



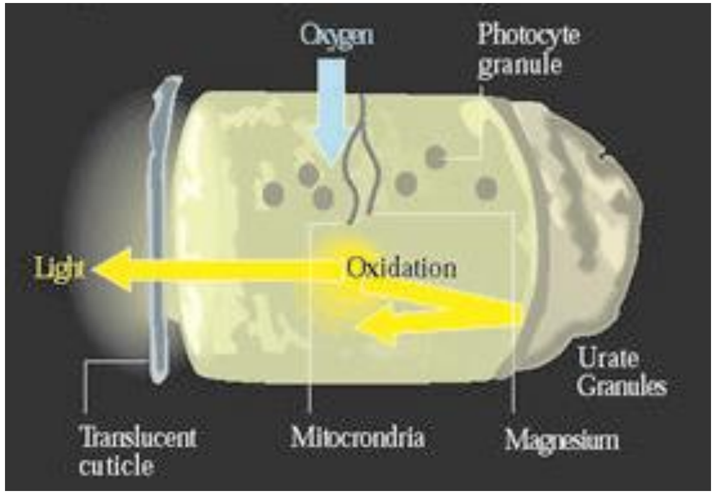


Converting chemical energy in light energy

Bioluminescence

Light energy is released by a chemical reaction involving the **luciferin** (a light-emitting pigment), and a **luciferase** (the enzyme component).

$$\text{Luciferin} + O_2 \xrightarrow[\text{other cofactors}]{\text{Luciferase}} \text{Oxyluciferin} + \text{light energy}$$

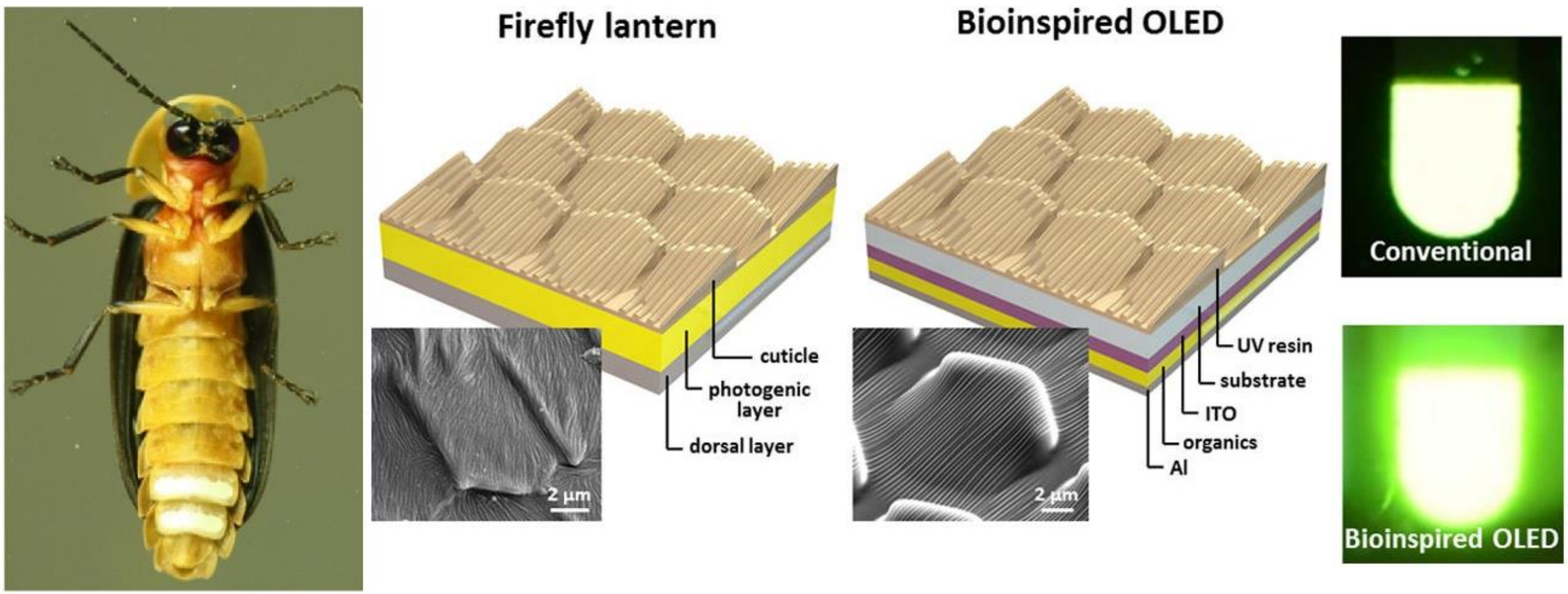


95% light
5% heat

10% light
90% heat

Converting chemical energy in light energy

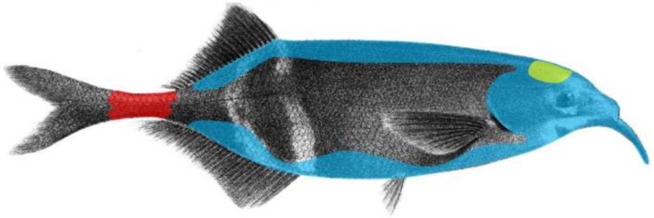
In a recent study, researchers have investigated the optical properties of the firefly's light-emitting cuticle, which is not smooth like most human-made lights, but instead is patterned with tiny hierarchical structures.
Inspired by these features, the researchers replicated the patterns to create a bioinspired organic light-emitting diode (OLED), resulting in a 60% increase in the light extraction efficiency and 15% wider angle of illumination.



Kim, J. J., Lee, J., Yang, S. P., Kim, H. G., Kweon, H. S., Yoo, S., & Jeong, K. H. (2016). Biologically inspired organic light-emitting diodes. Nano Letters, 16(5), 2994-3000.

The Electric Sense

Family: Mormyridae/Gymnarchidae



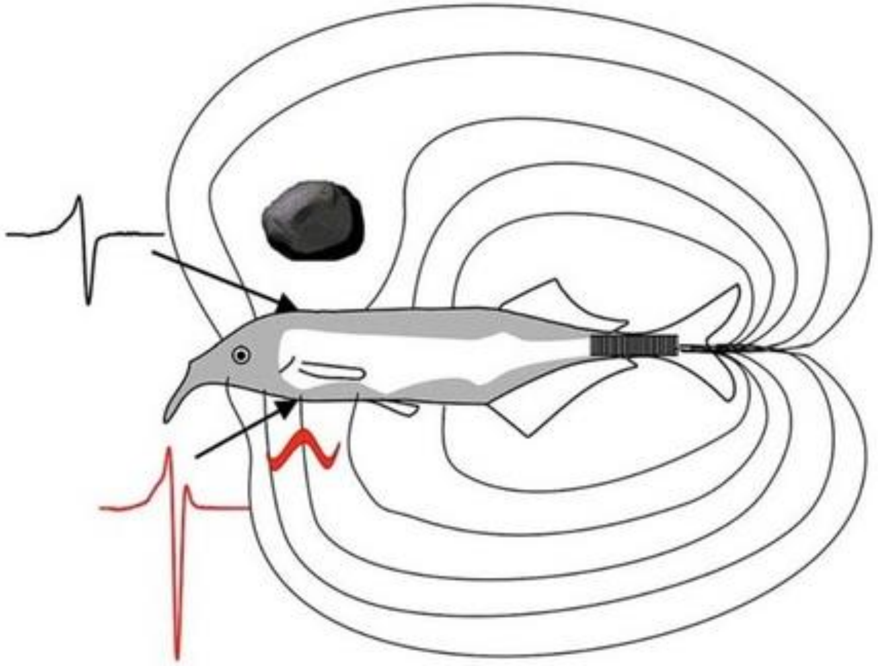
Electroreceptors

Central Nervous System

Electric Organ

Decreased field line density by a non-conductive object

Increased field line density by a worm (better conductor than the water)

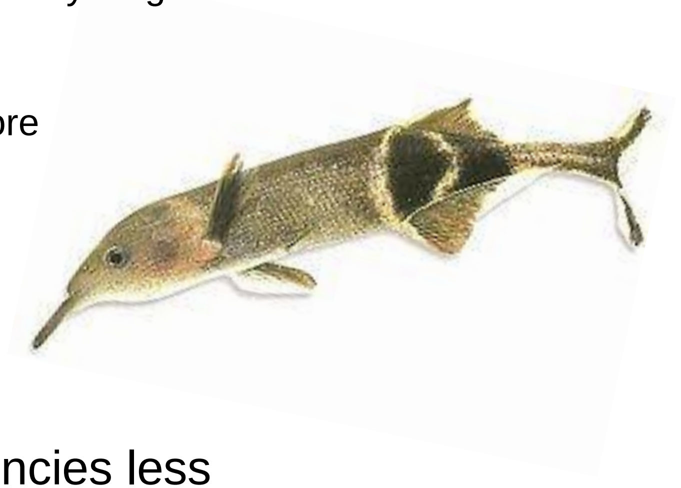


The Electric Sense

The **electric discharge** is produced by an electric organ that evolved from muscles located near the tail region

- A **pulse-type** electric organ discharge (EOD) is characterized by discrete EOD pulse separated by relatively long silent intervals much longer than the discharges;

- a **wave-type** EOD has its firing period and silence period approximately the same in length, and therefore a continuous signal with quasi-sinusoidal waveform (around 500 Hz) is formed.

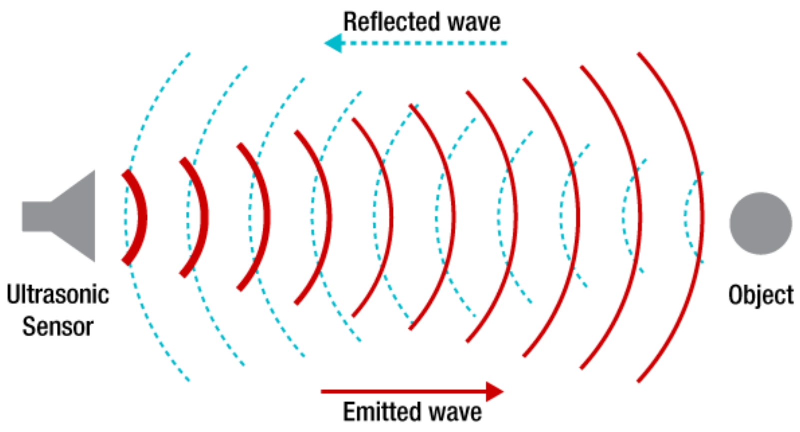


Electroreceptors

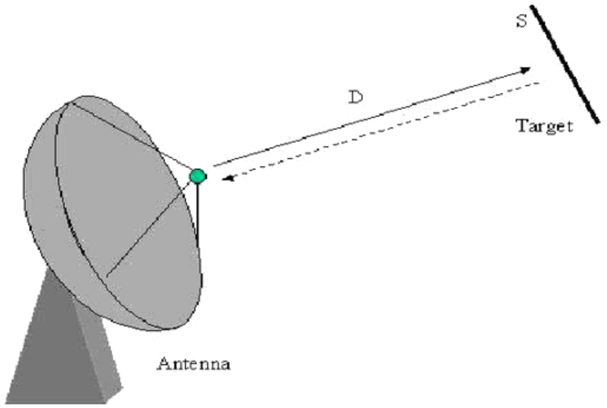
Electrolocation/ Locate preys → 0.01 mV/cm; sensitive to low frequencies less than 50 Hz

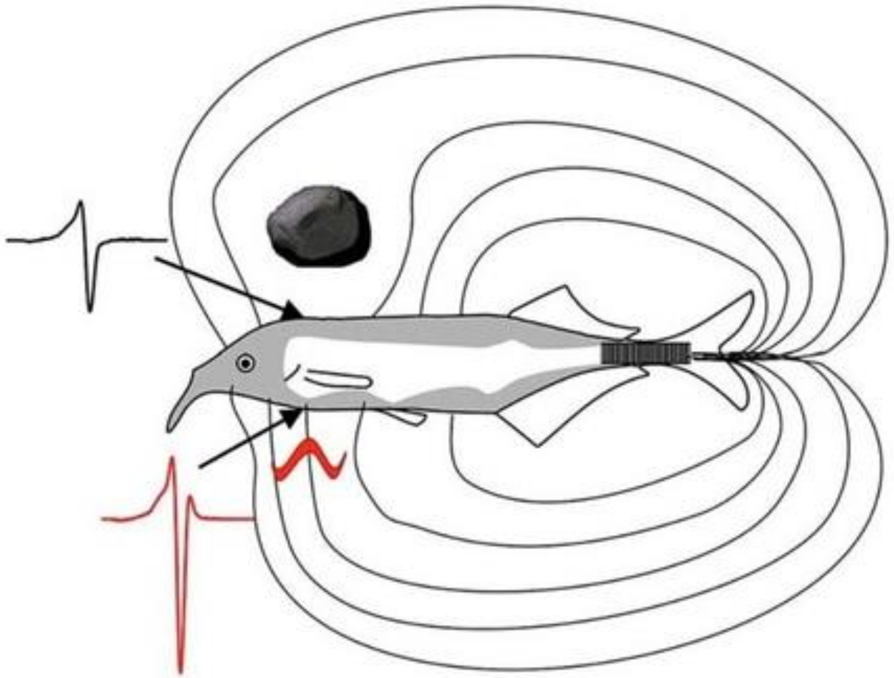
Electrocommunication → 0.1 mV to 10 mV/cm; Tens of Hz to more than 1 kHz

The Electric Sense



delayed reception





immediate reception